Basic Statistical Measures

Exam code: 9MA0

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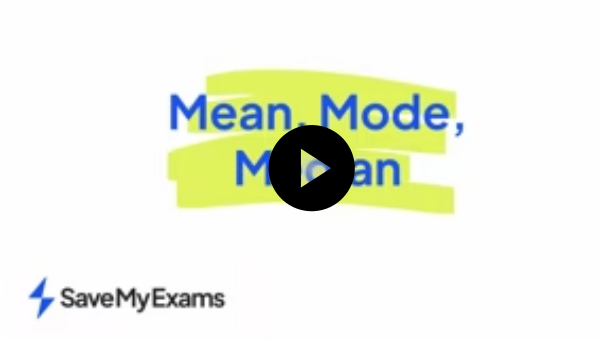

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Mean, mode and median

What are the mean, mode and median?

- Mean, mode and median are measures of **location**
  - A measure of location gives information about where data is in the number system
  - Mean, mode and median are **measures of central tendency**
  - They describe where the centre of the data is
- They are all types of **averages**
  - In Statistics it is important to be specific about which average you are referring to

How are the mean, mode and median calculated?

- You should already be familiar with finding the mean, median and mode from raw, ungrouped data
- The **mode** is the value that occurs **most often** in a data set
  - In a frequency table the **group** or **class** that occurs most often will be referred to as the **modal class**
  - A data set with more than one mode is **bimodal**
- The **median** is the **middle** value when the data is in **order of size**
  - If there are two values in the middle of the data set, the median is the **midpoint** of the two values

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- You may have to use **linear interpolation** when finding the median from a **grouped frequency table**
- The **mean** is the **sum** of all the values **divided by the number of values** in the data set

What are summary statistics and their notation?

- Summary statistics are information that summarises a set of data values
- For n items in a data set:
  - The sum of the data is represented by

$$\sum_{i=1}^{n} x_i = x_1 + x_2 + \ldots + x_n$$

- This is usually written $\sum x$ and reads as 'sigma x'

- The mean of the data is represented by

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} = \frac{\Sigma x}{n}$$

- This reads as 'x bar'
- You will come across more summary statistics later in the course

How do we choose the best measure of central tendency?

- It is often better to use one of the averages over the others, depending on the data set
  - It's a good idea to be aware of the advantages and disadvantages of the use of each average

- The mean uses all of the data values, this is good for a large data set where all of the values are close together, but also means that the mean can be affected by extreme values
- The median is not affected by very high or low values so is a good average to use in data sets with extreme values
- The mode is very useful in a lot of practical situations, however often there may be more than one mode, no mode or even a mode that is nowhere near the middle of the data set

Worked Example

For the data set given below, find the mode, median and mean.

23 19 14 28 27 19

Answer:

To find the median the data must be in order first

14 19 19 23 27 28

Median is half way between the two middle numbers.

$$\text{Median} = \frac{19 + 23}{2} = 21$$

Median = 21

The mean is the sum of the numbers divided by how many there are.

$$\text{Mean} = \frac{\Sigma x}{n} = \frac{14 + 19 + 19 + 23 + 27 + 28}{6} = \frac{130}{6}$$

Mean = 21.7 (3.s.f.)

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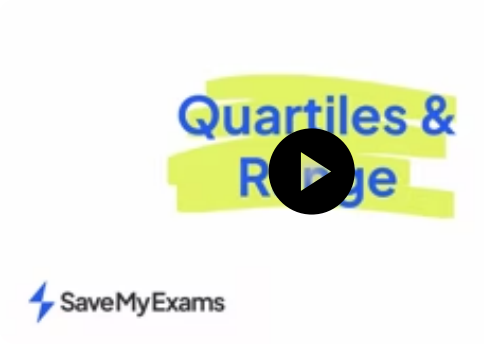

Did this video help you? Yes No

Quartiles & range

What are quartiles and percentiles?

- **Quartiles and percentiles** are measures of location
- Quartiles divide a population or data set into **four equal sections**
  - The **lower quartile, Q_1,** splits the lowest 25% from the highest 75%
  - The **median, Q_2,** is the value that is 50% of the way through the data
  - The **upper quartile, Q_3,** splits the lowest 75% from the highest 25%
- Percentiles divide the data into 100 parts
  - The 70th percentile lies seven-tenths of the way through the data
    - 70% of the data is below it and 30% is above it

How are quartiles calculated?

- For a data set of size, n,
  - To find the lower quartile, calculate $\frac{n}{4}$
    - If $\frac{n}{4}$ is an integer then the lower quartile is the midpoint of the corresponding value and the one above it
    - If $\frac{n}{4}$ is not an integer then the lower quartile is the value corresponding to the next integer up

midpoint of the corresponding value and the one above it
    - If $\frac{3n}{4}$ is not an integer then the upper quartile is the value corresponding to the next integer up
- You can also use your calculator to find the quartiles, make sure you know how to put your calculator into STAT mode, enter the data and find the values of Q_1, Q_2 and Q_3
  - You could be expected to be able to do this in the exam

What are the range and interquartile range?

- The **range** and **interquartile range** are both **measures of spread**
- A measure of spread gives information about how spread out the data set is
- The **range** is the difference between the **largest** and **smallest** values in the data set
  - All data points in the set will be included in the range, including **extreme values**

- The **interquartile range** is the difference between the **upper quartile** and the **lower quartile**
  - Only the middle 50% of the data is included in the interquartile range
  - It is not affected by extreme values

- The **units** for range and interquartile range are the same as the units for the original data

What is the interpercentile range?

- Sometimes an **interpercentile range** could be asked for
  - this is the difference between two given percentiles
    - For example, the 20th to 80th interpercentile range would be the difference between the 80th percentile and the 20th percentile

- The **units** for the percentiles and interpercentile range are the same as the units for the original data

Worked Example

Find the range and interquartile range for the data set given below

43 29 70 31 84 56 17

Answer:

Examiner Tips and Tricks

Be aware of the difference between averages and ranges, especially when answering contextual questions asking you to describe or compare data. Remember, averages give an indication of where the data are whilst range gives an indication of how varied the data are.

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